



HK IMO Training 2023-2024
Problem Set 12
For 6 January 2024



Problem 45. Let D be a point on the side BC of $\triangle ABC$ such that the tangent at A to the circumcircle of $\triangle ACD$ bisects BD . Prove that $\sqrt{2}BC \geq AB + AC$.

Problem 46. There are 33 students in a class. Each student writes down two numbers on the blackboard. The first number indicates the number of other students in the class who have the same surname as this student, and the second number indicates the number of other students in the class who have the same first name as this student. After all students have written down the numbers, it is found that each of the numbers $0, 1, 2, \dots, 10$ appears at least once on the blackboard. Prove that there are two students in the class who have the same surname and the same first name.

Problem 47. For each positive integer n , let $\varphi(n)$ be Euler's totient function, and let $\pi(n)$ be the number of prime numbers less than or equal to n . Prove that

$$\pi(m) - \pi(n) \leq \frac{(m-1)\varphi(n)}{n}$$

for all positive integers m and n . When does equality hold?

Problem 48. Two sequences (a_n) and (b_n) are defined as follows:

$$\begin{aligned} a_0 &= 0, & a_1 &= 1, & a_{n+1} &= \frac{2018}{n}a_n + a_{n-1}, \\ b_0 &= 0, & b_1 &= 1, & b_{n+1} &= \frac{2020}{n}b_n + b_{n-1} \end{aligned}$$

for all $n \geq 1$. Prove that $\frac{a_{1010}}{1010} = \frac{b_{1009}}{1009}$.